

Conceptual Architecture for Normative AI Agents

A methods paper on the encoding of political-ethical concepts for agent-based simulation

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Abstract

This paper presents a framework for encoding canonical definitions of five political-ethical concepts — freedom, justice, authority, care, and loyalty — into transparent, uncertainty-aware parameter representations that generate distinct and interpretable social dynamics in an agent-based simulation. Each concept is reduced to a single canonical definition grounded in the psychological literature: Self-Determination Theory and Psychological Reactance Theory for freedom; Equity Theory and the organisational justice framework for justice; Milgram's obedience paradigm together with Kelman's three-process model for authority; the Interpersonal Reactivity Index and empathy-altruism literature for care; and identity-fusion theory for loyalty. These definitions are encoded as probability distributions over a shared set of ten parameters, each defined on a continuous 0–1 scale with explicit conceptual endpoints. Agents are sampled from a ten-dimensional joint distribution constructed via a Gaussian copula with empirically calibrated Beta marginals, and are placed into one of thirty-two possible societal configurations that define the normative architecture of the simulation environment. Resulting dynamics are evaluated against a set of five behavioural benchmarks — one per concept — drawn from canonical social psychology: Milgram for authority, Asch for loyalty, the Ultimatum Game for justice, bystander helping for care, and reactance restoration for freedom. The five-concept pentad is justified through a convergence criterion: each concept appears in at least three of four independently-developed psychological frameworks (Rokeach 1973, Duckitt & Sibley 2009/2010, Schwartz 1992/2012, Moral Foundations Theory per Atari et al. 2023). The central open problems — the operationalisation of moral performance, the empirical grounding of parameter distributions, and the validation of the joint dependency structure — are identified explicitly. The framework contributes a methodology for treating normative reasoning as a structural problem of conceptual encoding rather than a compliance problem addressed through external constraint.

1. Motivation and Research Context

AI systems increasingly mediate consequential decisions — in work, governance, communication, and care. Their behaviour is shaped by engineering interventions that have grown in sophistication: prompt engineering gave way to context engineering, which in turn has given way to harness engineering. As these systems approach broader forms of agency, a further engineering layer becomes load-bearing: the engineering of the normative frames through which these systems interpret actions, directives, and outcomes.

Current approaches to normative reasoning in AI systems rely predominantly on external rule patches: alignment training, constitutional constraints, and post-hoc safety filters. These approaches treat ethical behaviour as a compliance problem rather than a structural one. This paper proceeds from a different premise: that the normative interpretation of an AI system could be shaped by an intrinsic conceptual architecture — a formal, explicit encoding of the political and ethical concepts that structure its judgment from the inside rather than constraining it from the outside.

The immediate research question is deliberately narrow: **can canonical definitions of five political-ethical concepts be encoded into uncertainty-aware parameter profiles that generate distinguishable and interpretable social dynamics?** If the answer is yes, even in a constrained proof-of-concept, it suggests that structured conceptual encodings can do genuine explanatory work rather than serving as decorative philosophical preface.

The framework stands outside any single ideological tradition, seeking instead to formalise and compare the structural assumptions embedded in each. Its scope is limited to the methodological architecture and execution plan. Implementation, empirical validation, and operationalisation of key terms remain open problems,

acknowledged explicitly throughout. The work does not claim to solve alignment, produce a fully ethical artificial agent, or construct a universal political ontology. Its contribution, if successful, is a methodology for treating normative reasoning as structural encoding rather than external constraint.

2. The Five Concepts

Rather than attempting to model all of political theory at once, the framework starts from five political-ethical concepts that recurrently structure disagreement about institutions and collective life: freedom, justice, authority, care, and loyalty. The choice is not arbitrary.

2.1 Selection criterion

Three independently-developed frameworks in political psychology, together with a fourth in moral psychology, converge on dimensions that correspond closely to these five concepts. The selection rule is falsifiable: each concept must appear in at least three of the four frameworks listed below.

- **Rokeach (1973)** two-value model: freedom and equality as orthogonal axes differentiating all major political ideologies. Anchors freedom directly and justice via equality. Replicated and extended by Braithwaite (1994) and Cochrane (1979), with the caveat that the freedom axis is rhetorically universal and therefore less discriminating than equality across real-world party data.
- **Duckitt & Sibley (2009, 2010)** dual-process model: RWA (authoritarianism, authority-submission) and SDO (preference for group hierarchy, inverse of egalitarian justice) as the two empirically orthogonal dimensions of ideological attitudes. Validated across decades of factor-analytic work (Sibley & Duckitt 2008 meta-analysis). The opposite pole of RWA is consistently labelled "personal freedom" in the literature, giving authority, justice, and freedom anchors in a single framework.
- **Schwartz (1992, 2012)** value circumplex: ten motivationally distinct values arranged in a circular structure around two orthogonal axes (self-enhancement/transcendence, openness/conservation). Validated across 80+ countries (Rudnev et al. 2018). Places self-direction (freedom-adjacent), power (authority-adjacent), universalism (justice-adjacent), benevolence (care-adjacent), and conformity/tradition (loyalty-adjacent) as distinct motivational clusters with built-in orthogonality from the circular structure.
- **Moral Foundations Theory, Atari et al. (2023) reformulation** — six foundations: Care/harm, Equality, Proportionality, Authority/respect, Loyalty/betrayal, Sanctity, Liberty/oppression. The 2023 reformulation splits the Fairness foundation into Equality and Proportionality, which together map onto justice in the sense used here.

Freedom, justice, and authority appear in all four frameworks. Care and loyalty appear in three of four. The remaining candidates (tradition, sanctity, security, achievement) appear in two or fewer, or are redundant with parameters already in the encoding. The pentad is therefore the set of political-ethical concepts for which there is converging cross-tradition psychological support under a falsifiable selection rule.

Each concept is reduced to a single canonical definition grounded in the psychological rather than philosophical literature. The rationale is that the simulation ultimately models behaviour — how agents act under constraint, evaluate outcomes, respond to directives, extend concern, and enact commitment — and the psychological literature provides definitions closer to the mechanisms the simulation must encode. The philosophical traditions remain as background context, but the definitions that enter the encoding pipeline are psychological.

2.2 Freedom

Freedom is defined as the subjective experience of volitional agency — the perception that one's actions are self-endorsed and congruent with one's integrated values — coupled with a motivational disposition to resist perceived threats to that agency.

This definition synthesises two major psychological frameworks. The positive dimension — freedom as experienced autonomy — draws on Self-Determination Theory (Ryan & Deci, 1985; 2000), which posits three innate psychological needs (autonomy, competence, relatedness) and defines autonomy not as independence

from others but as acting in accordance with one's integrated self rather than under external pressures. SDT provides a continuum from fully extrinsic to fully intrinsic motivation, with intermediate stages of internalisation, giving the encoding a natural distribution shape rather than a binary free/unfree distinction. SDT has demonstrated reasonable cross-cultural validity (Chirkov, Ryan, Kim & Kaplan, 2003), which is relevant for distribution modelling.

The reactive dimension — freedom as resistance to constraint — draws on Psychological Reactance Theory (Brehm, 1966; Brehm & Brehm, 1981). PRT defines reactance as an unpleasant motivational arousal that emerges when people experience a threat to or loss of their free behaviours, serving as a motivator to restore freedom. The amount of reactance depends on the importance of the threatened freedom and the perceived magnitude of the threat. Where SDT describes what freedom feels like when satisfied, Brehm describes what happens when it is threatened — and the behavioural consequences (boomerang effects, source derogation, hostile affect) map directly to agent behaviour in the simulation.

2.3 Justice

Justice is defined as the cognitive-affective appraisal of proportionality between contributions and outcomes — evaluated through social comparison, sensitive to both distributional and procedural dimensions, and generating corrective motivation when perceived imbalance exceeds a subjective threshold.

This definition integrates three streams of psychological research. The foundational mechanism is Adams' Equity Theory (1963; 1965), which posits that individuals assess fairness by comparing their input/output ratios against those of referent others, with perceived imbalance generating distress and corrective behaviour. The evaluation of equity is socially dependent and entails complex psychological and cognitive processes — it is not merely a calculation of objective reward but a subjective, comparison-driven appraisal.

The distributional/procedural distinction draws on the organisational justice framework: Thibaut and Walker (1975) on procedural justice, Leventhal (1980) on allocation preferences, and Colquitt's (2001) four-factor validation. This tradition demonstrates that people evaluate not only whether outcomes are fair but whether the processes that produced them are fair — and that these two evaluations are psychologically separable, with different predictive validity for downstream behaviour (commitment, retaliation, compliance). Tyler and Blader's (2003) group engagement model extends this into the authority-legitimacy domain, showing that procedural justice shapes cooperation through its effect on social identity.

The variation in what "fairness" means across populations draws on Moral Foundations Theory (Haidt & Graham, 2007), specifically the Atari et al. (2023) reformulation, which splits the Fairness foundation into Equality ("intuitions about equal treatment and equal outcome for individuals") and Proportionality ("intuitions about individuals getting rewarded in proportion to their merit or contribution"). This directly informs how the justice parameters should be distributed across a simulated population.

2.4 Authority

Authority is defined as the perceived legitimacy of asymmetric social influence — the degree to which an agent attributes to another the right to prescribe behaviour — mediated by mechanisms of compliance (instrumental), identification (relational), and internalisation (value-congruent), and modulated by contextual cues of expertise, institutional role, and proximity.

The empirical backbone is Milgram's obedience paradigm (1963; 1974). Across 23 conditions with 636 participants, Milgram demonstrated that obedience to authority is modulated by at least eight factors: the experimenter's directiveness, legitimacy, and consistency; group pressure on the subject to disobey; and the indirectness, proximity, and intimacy of the relation between subject and victim, as well as the distance between subject and experimenter (Haslam, Loughnan & Perry, 2014, meta-synthesis). The 65% baseline obedience rate, the dramatic variation across conditions (from 92.5% under diffused responsibility to near-zero under peer rebellion), and the robustness across replications and decades provide a behavioural landscape against which the simulation's authority encoding can be directly validated.

The mechanistic model is Kelman's three-process framework (1958; 1974), which distinguishes compliance (behavioural conformity driven by reward/punishment, dissolving when surveillance ends), identification (conformity driven by desire to maintain a valued relationship with the authority source, dissolving when the relationship ends), and internalisation (acceptance because the authority's prescriptions align with one's own values, persisting independently). These three processes predict qualitatively different behavioural patterns and map directly onto the simulation's Internalization Dependence and Legitimacy Locus parameters.

Moral Foundations Theory's Authority/respect foundation (Atari et al., 2023) provides evolutionary-psychological grounding: authority sensitivity varies systematically across populations and correlates with political orientation, informing distribution modelling. Tyler (2006) provides the legitimacy-as-psychological-construct anchor that connects authority to procedural justice at the parameter level.

2.5 Care

Care is defined as other-oriented concern for the welfare of those perceived as morally considerable — extended through affective mechanisms (empathic concern, felt compassion) and cognitive mechanisms (perspective-taking, reasoned beneficence), activated by cues of suffering or need, and modulated by the perceived breadth of the moral circle and the social-relational embedding of the potential recipient.

The definition integrates three streams of research. The mechanistic backbone is Davis's (1983) Interpersonal Reactivity Index, which empirically distinguishes affective empathy (Empathic Concern — felt sympathy for unfortunate others; Personal Distress — self-oriented unease in tense settings) from cognitive empathy (Perspective Taking — spontaneous adoption of another's point of view; Fantasy — imaginative projection into fictional others). The IRI has become the standard dispositional-empathy instrument, cited over ten thousand times, and the EC/PT subscale distinction directly anchors the framework's Affective Weighting parameter in the care domain.

The motivational mechanism is Batson's empathy-altruism hypothesis (1981; 2011 synthesis): induced empathy produces genuinely altruistic helping, distinct from egoistic distress-reduction. Batson's laboratory experiments give quantified behavioural outputs under manipulated empathy and cost conditions. While the debate with Cialdini et al. (1997) on "oneness" as a confounding mediator remains live, the mainstream position supports the empathy-altruism dissociation at the behavioural level.

The breadth dimension — how far care is extended — draws on Crimston, Bain, Hornsey and Bastian's (2016) Moral Expansiveness Scale, which operationalises Singer's (1981) expanding-circle concept. The MES measures the range of entities (human and non-human) deemed worthy of moral concern, with stable correlations to identification-with-all-humanity, universalism values, and connectedness to nature (MESx validation, Crimston et al. 2018). The MES is the primary calibration anchor for the Moral Scope parameter and provides the direct instrument for the care-scope mapping.

The activation threshold — how much suffering must register before care translates into helping behaviour — draws on the bystander-effect literature (Latané & Darley, 1968; 1970; Fischer et al., 2011 meta-analysis of 53 studies). This literature provides quantified baselines for the activation of helping under varying bystander-presence and danger conditions, and serves as the behavioural benchmark for the care concept.

2.6 Loyalty

Loyalty is defined as sustained in-group commitment extending beyond instrumental calculation — the disposition to maintain pro-group behaviour at personal cost, mediated by identification with group identity and, at its extreme, by fusion of personal and social identities — modulated by contextual cues of in-group legitimacy, ritual enactment, and perceived threat to the group.

The empirical backbone is identity fusion theory (Swann, Jetten, Gómez, Whitehouse & Bastian, 2012), which distinguishes fusion — a visceral sense of oneness with the group in which personal and social identities become porous — from ordinary group identification. Fusion theory was originally developed to explain extreme pro-group behaviour in cases of terrorism and self-sacrifice, but the construct generalises. A 2023 meta-analysis of 90 studies across 55 reports (N = 36,880) confirmed a strong, heterogeneous relationship between fusion and extreme pro-

group orientations (Varmann et al., 2023). The 2024 revised theory (Swann et al., 2024 CIFT) provides the current instrument and framework.

The mechanistic ladder is Kelman's (1958) compliance–identification–internalisation framework, originally formulated for authority but applying with equal force to loyalty: identification as relationship-maintained conformity dissolves when the relationship ends; internalisation as value-congruent commitment persists independently. Identity fusion extends the internalisation pole into a state where the group is not merely endorsed but constitutive of the self.

The ritual-enactment dimension — how loyalty is maintained through procedural performance rather than substantive fidelity — draws on Whitehouse's (2018) distinction between imagistic (high-affect, infrequent, transformative) and doctrinal (routinised, frequent, low-affect) modes of religious/communal commitment. This anchors the Procedural Dependence parameter in the loyalty domain. Atran and Ginges (2012) on sacred values provides a complementary framework for loyalty commitments that resist cost-benefit framing.

The scope dimension — the size of the loyalty target — draws on Haidt's (2012) Loyalty/binding foundation and McFarland et al.'s (2012) Identification with All Humanity scale, which demonstrates that loyalty objects range from immediate kin through nation to humanity itself. Conjunction with the Moral Scope parameter produces the tension between particularistic and universalist loyalty that is a central feature of the framework's configuration space.

3. Encoding Architecture

3.1 Shared parameter set

A deliberate design choice is made to use a single shared parameter set across all five concepts rather than topic-specific parameters. Topic-specific parameters would multiply complexity and preclude the cross-concept comparison that is the research target. The universality claim is load-bearing: if any parameter fails to apply non-trivially to all five concepts, the architecture fails and the system must decompose into five parallel encoding pipelines. The ten parameters below survive this test. Each applies to each concept with a documented behavioural claim and at least one literature anchor; the full ten-by-five mapping (fifty cells, eighty-four citations) is provided in the companion Parameter-to-Concept Mapping Matrix.

The parameter set was developed iteratively and validated for non-redundancy through pairwise endpoint analysis, confirming that all forty-five possible parameter pairings produce genuinely distinct 2×2 behavioural profiles. Each parameter is defined on a continuous [0, 1] scale with explicit conceptual endpoints.

1. Legitimacy Locus — on what grounds is a norm, action, outcome, directive, care-claim, or loyalty-claim treated as valid? 0 = validity depends on internal endorsement, personal judgment, or self-authentication. 1 = validity depends on external recognition, shared norms, role structure, or institutional warrant. Primary anchor: Deci & Ryan GCOS autonomy-orientation (1985). Maps across concepts: freedom asks whether agency is validated internally or by recognised conditions; justice asks whether fairness judgments are personal-intuitive or norm-anchored; authority asks whether obedience is earned by office, expertise, or shared norm; care asks whether compassion extends on personal grounds or by institutional deservingness; loyalty asks whether group bonds are sustained by internal commitment or external markers.

2. Constraint Sensitivity — how readily does the agent interpret influence, asymmetry, or outcome-shaping as meaningful restriction? 0 = low sensitivity; influence and structure are registered as environmental features. 1 = high sensitivity; even soft nudges register as potential constraints. Primary anchor: Hong Psychological Reactance Scale (Hong & Faedda 1996). Central to freedom but determining how readily injustice is detected, whether authority is experienced as coercive, whether help is received as paternalism, and whether in-group expectations feel like identity-coercion.

3. Response Threshold — how much deviation, imbalance, or illegitimacy is tolerated before the agent is moved to respond? 0 = high tolerance; responds only to major violations. 1 = low tolerance; activated by mild deviations. Primary anchor: Ultimatum Game rejection thresholds (Wallace et al. 2007). Determines when constraint triggers

resistance (freedom), when inequity triggers correction (justice), when illegitimate command triggers refusal (authority), when suffering activates helping (care), and when group-threat activates protective action (loyalty).

4. Mode of Response — when the concept is activated, what form does response take? 0 = internal, reflective, self-adjusting; the self adjusts rather than the world. 1 = external, behavioural, confrontational; the world is acted on rather than the self. Primary anchors: STAXI anger-in/anger-out (Spielberger 1988), Thomas-Kilmann conflict styles, Davis IRI Empathic Concern vs. Personal Distress distinction, Hirschman's exit-voice-loyalty framework (1970). Bridges the parameter layer to observable behaviour across all five concepts.

5. Relational Embedding — to what extent is the concept interpreted through social roles, proximity, and interpersonal bonds rather than abstract standalone agents? 0 = atomised, agent-centred, abstract-person model. 1 = role-sensitive, relational, socially embedded. Primary anchor: Singelis Self-Construal Scale (1994). Foundational for loyalty — loyalty presupposes relational embedding — and load-bearing for care through the universalist-vs-partialist distinction (Bloom 2016; Greene 2013).

6. Procedural Dependence — how much does the concept depend on how something is done rather than only what outcome occurs? 0 = outcome-dominant; results matter, methods are secondary. 1 = process-dominant; fair procedure matters independently of outcomes. Primary anchors: Colquitt's Organisational Justice Scale (2001), Tyler & Blader group engagement model (2003). Empirically separable from distributive judgment; authority legitimacy depends heavily on mandate, due process, and consent; loyalty can be enacted through ritual form (Whitehouse 2018) or substantive support.

7. Tolerance for Asymmetry — how much unequal influence, rank, entitlement, or differential standing is acceptable? 0 = asymmetry is inherently suspect, requiring justification; default expectation is symmetry. 1 = asymmetry is accepted if intelligible or justified; default accommodates hierarchy. Primary anchor: Social Dominance Orientation (Pratto et al. 1994; Ho et al. 2015 SDO7 update). The best-empirically-anchored parameter. Duckitt & Sibley's dual-process model (2009, 2010) extends this through the RWA-SDO structure.

8. Internalization Dependence — how much does realisation of the concept depend on inward endorsement rather than external enforcement alone? 0 = can operate through compliance, sanctions, and surface behaviour alone. 1 = requires genuine belief, endorsement, or value-congruence. Primary anchor: Self-Regulation Questionnaire (Ryan & Connell 1989). Directly operationalises Kelman's compliance–identification–internalisation ladder and connects all five concept definitions at the deepest level.

9. Moral Scope — how broadly is the concept extended across persons and cases? 0 = local, role-bound, partial, context-limited. 1 = universalised, generalisable, broadly applied. Primary anchor: Moral Expansiveness Scale (Crimston et al. 2016; MESx short-form Crimston et al. 2018). Secondary: Identification with All Humanity (McFarland et al. 2012). Replaces the earlier MFQ-only anchoring; MES is the direct instrument for expanding-circle operationalisation. Load-bearing for care (who is cared for) and loyalty (how wide the loyalty target), and in productive tension with Tolerance for Asymmetry.

10. Affective Weighting — the extent to which the concept is processed through emotional engagement versus cognitive deliberation. 0 = cognitive, deliberative, reasoned. 1 = affective, intuitive, felt. Primary anchor: Davis IRI Empathic Concern vs. Perspective Taking subscale ratio (1983). Supporting: Greene et al. (2001) dual-process moral judgment; Haidt (2001) social intuitionist model; Decety & Jackson (2004) affective vs. cognitive empathy. Cuts across all five concepts in a distinct way — felt autonomy vs. reasoned autonomy, moral outrage vs. equity calculation, charismatic vs. rational-legal authority, empathic concern vs. reasoned beneficence, identity fusion vs. deliberated commitment.

The parameter set is theory-driven and developed from close reading of the canonical definitions. It is not treated as an oracle that automatically discovers the ontology of political concepts. Exploratory dimensional reduction may later test whether any parameters are empirically redundant despite conceptual distinctness; the mapping matrix provides the concept-by-parameter cross-section that would be used for that analysis.

3.2 Coding protocol

Each concept's canonical definition is encoded as a probability distribution over the shared parameters rather than a single fixed score. A transparent rubric specifies what high and low values mean for each axis — the endpoint

definitions above serve as this rubric. Ambiguous aspects of the definition remain broad; clearer aspects narrow. The aim is not to automate philosophy away, but to make interpretive judgment explicit, comparable, and uncertainty-aware.

3.3 Distribution modelling

The distributions assigned to each parameter must reflect real human populations, not arbitrary priors. The target population for the proof-of-concept is adults in consolidated Western democracies (United States, United Kingdom, Western Europe, Scandinavia, Australia), with Italian and Southern European data as primary calibration anchors where available. Anglophone data fills gaps. This scope ensures instrument validity across all proxy measures while avoiding the monocultural trap of US-only calibration. The choice of Western democracies as reference class also enables the Milgram retrodiction test: the same population scope that generated the empirical benchmarks is the one against which the simulation's distributions are calibrated.

The natural distribution family for bounded continuous data on $[0, 1]$ is the Beta distribution, parameterised by two shape parameters (α, β). The Beta family can take virtually any shape on the unit interval — uniform, U-shaped, J-shaped, symmetric bell, left-skewed, right-skewed — making it the default candidate for all ten parameters. The mean is given by $\alpha/(\alpha+\beta)$, and the concentration ($\alpha+\beta$) controls variance: higher sums produce tighter distributions. For parameters where empirical data reveals genuine bimodality — notably Tolerance for Asymmetry (parameter 7) and Moral Scope (parameter 9), where political polarisation may produce bimodal score distributions — a mixture of two Beta distributions serves as the documented fallback. The copula procedure described in Section 3.4 works identically with mixture marginals: Step 3 uses the mixture CDF inverse instead of a single Beta inverse, requiring no architectural change.

For each parameter, the distribution shape is calibrated against empirical data from a proxy instrument whose score distributions provide calibration anchors. The mapping from empirical instruments to framework parameters is not one-to-one — no existing scale measures exactly "Response Threshold" or "Relational Embedding." The argument is that if a known instrument measuring a closely related construct produces distribution shape X in population Y, the corresponding parameter distribution should approximate that shape after appropriate rescaling to $[0, 1]$. Visible density plots for each parameter are provided in the companion distributions document.

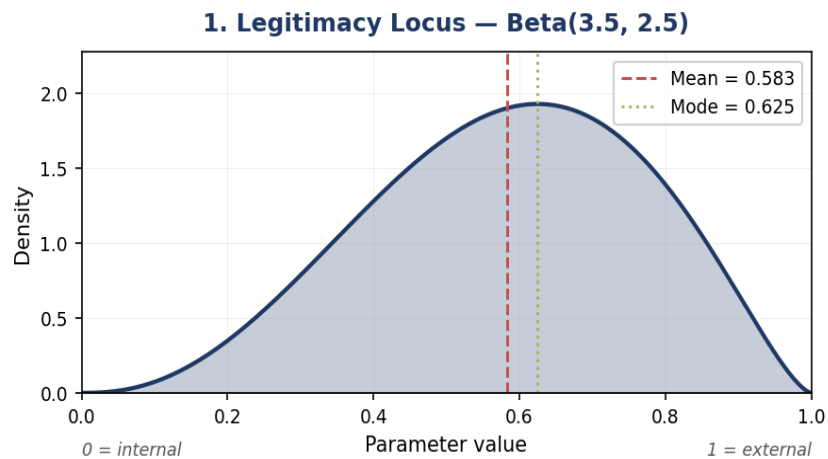
Table 1. Marginal Beta distributions for the ten agent parameters, calibrated to Western democratic adult populations.

#	Parameter	Scale	α	β	Mean	Skew	Proxy Instrument
1	Legitimacy Locus	0=internal, 1=external	3.5	2.5	0.58	Internal-leaning	GCOS (Deci & Ryan, 1985)
2	Constraint Sensitivity	0=low, 1=high	2.5	2.0	0.56	Slight high	HPRS (Hong & Faedda, 1996)
3	Response Threshold	0=tolerant, 1=hair-trigger	2.5	2.5	0.50	Symmetric	UG rejection (Wallace et al. 2007)
4	Mode of Response	0=internal, 1=external	2.0	2.5	0.44	Internal	STAXI / Thomas-Kilmann / IRI
5	Relational Embedding	0=atomised, 1=relational	2.0	3.0	0.40	Atomised	Singelis SCS (1994)
6	Procedural Dependence	0=outcome, 1=process	2.5	2.0	0.56	Outcome-leaning	Colquitt (2001) / trolley
7	Tolerance for Asymmetry	0=egalitarian, 1=hierarchical	2.0	3.5	0.36	Egalitarian	SDO7 (Pratto et al. 1994; Ho et al. 2015)
8	Internalisation Dependence	0=surface, 1=endorsement	3.0	2.0	0.60	Endorsement	SRQ (Ryan & Connell, 1989)
9	Moral Scope	0=local, 1=universal	1.8	1.5	0.55	Flat/universal	MES (Crimston et al. 2016)

#	Parameter	Scale	α	β	Mean	Skew	Proxy Instrument
10	Affective Weighting	0=cognitive, 1=affective	2.2	2.5	0.47	Slight cognitive	Davis IRI — EC/PT (1983)

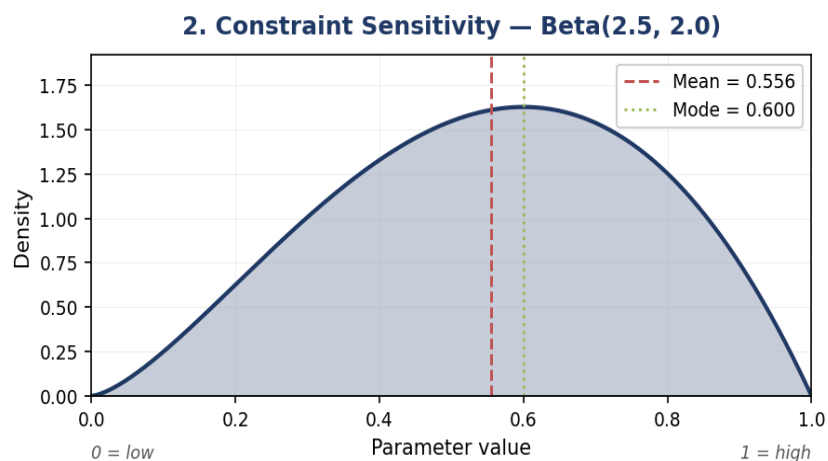
3.3.1 Legitimacy Locus — Beta(3.5, 2.5)

Proxy: General Causality Orientations Scale (GCOS; Deci & Ryan, 1985). Western adults consistently score higher on autonomy orientation than control orientation across US, European, and Australian samples. The autonomy subscale dominates in most respondents, producing a distribution skewed toward internal legitimacy. The GCOS has been validated across multiple populations with satisfactory internal consistency ($\alpha \approx .63$ –.75 across subscales).



3.3.2 Constraint Sensitivity — Beta(2.5, 2.0)

Proxy: Hong Psychological Reactance Scale (HPRS; Hong & Faedda, 1996). Refined on 3,085 respondents from metropolitan Sydney. Score distributions are moderately right-skewed: most individuals show moderate reactance with a tail of highly reactive persons. Confirmed in Finnish replications ($n = 624, 518$). The HPRS can be treated as unidimensional at the second order despite a four-factor first-order structure (Shen & Dillard, 2005).

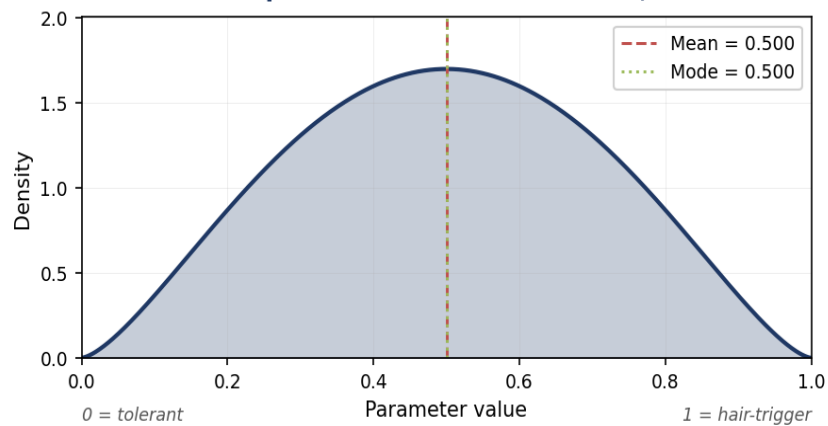


3.3.3 Response Threshold — Beta(2.5, 2.5)

Proxy: Ultimatum Game rejection thresholds. A Swedish Twin Registry study ($n = 653$) found a mean acceptance threshold of approximately 33% of the stake, with over 40% of variation explained by additive genetic effects (Wallace et al., 2007). A meta-analysis of 37 papers found that Western proposers offer approximately 40% on

average and 16% of offers are rejected overall (Oosterbeek et al., 2004). Individual thresholds show substantial heterogeneity, justifying a symmetric Beta distribution.

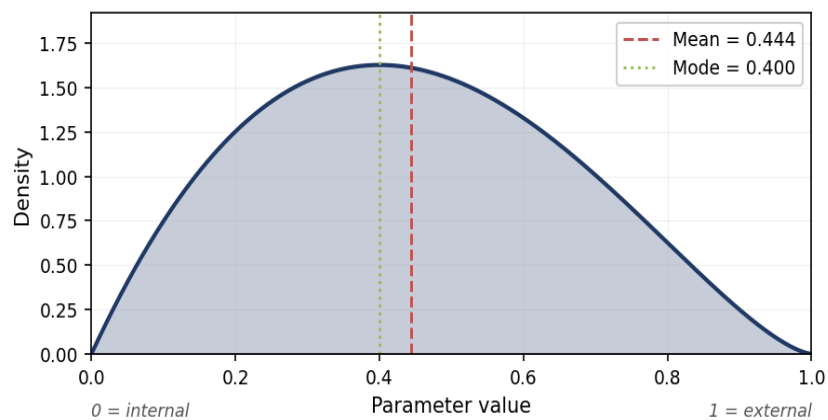
3. Response Threshold — Beta(2.5, 2.5)



3.3.4 Mode of Response — Beta(2.0, 2.5)

Proxies: State-Trait Anger Expression Inventory (STAXI; Spielberger, 1988), Thomas-Kilmann Conflict Mode Instrument, and Davis IRI Empathic Concern vs. Personal Distress distinction. The Anger-In vs. Anger-Out subscale ratio shows a slight lean toward internalisation in Western populations. Competing and confronting styles in Thomas-Kilmann are consistently less common than avoiding and accommodating in general population samples.

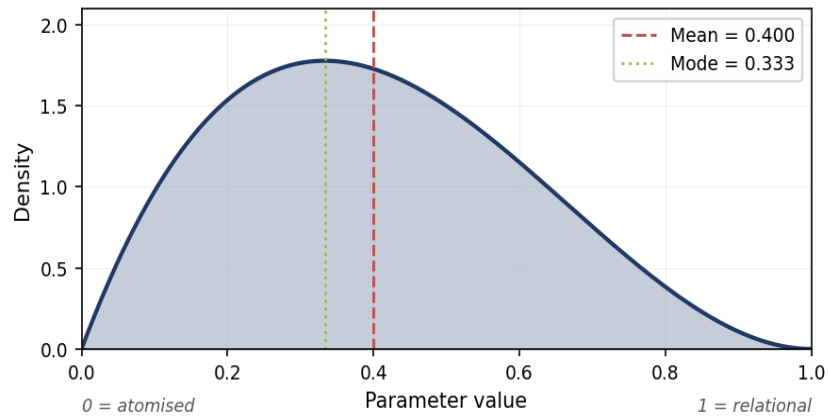
4. Mode of Response — Beta(2.0, 2.5)



3.3.5 Relational Embedding — Beta(2.0, 3.0)

Proxy: Self-Construal Scale (SCS; Singelis, 1994). Western populations reliably score higher on independent than interdependent self-construal. The two dimensions are orthogonal ($r = -.044$ in the original validation). Southern European populations show somewhat higher interdependence than Anglophone ones due to stronger family embeddedness structures. An Italian sub-calibration would shift toward Beta(2.5, 3.0).

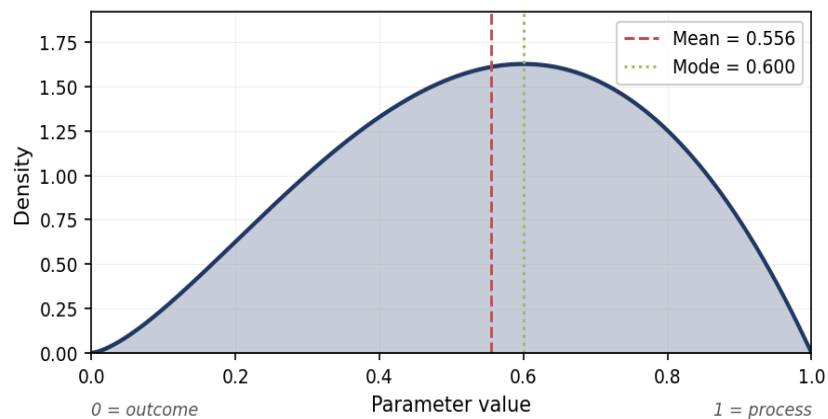
5. Relational Embedding — Beta(2.0, 3.0)



3.3.6 Procedural Dependence — Beta(2.5, 2.0)

Proxy: Colquitt's Organisational Justice Scale (2001), specifically the procedural-to-distributive justice sensitivity ratio, combined with trolley-problem response data. Western populations split approximately 60/40 in favour of consequentialist over deontological reasoning. Distributive concerns tend to be slightly stronger predictors of satisfaction in Western workplace samples.

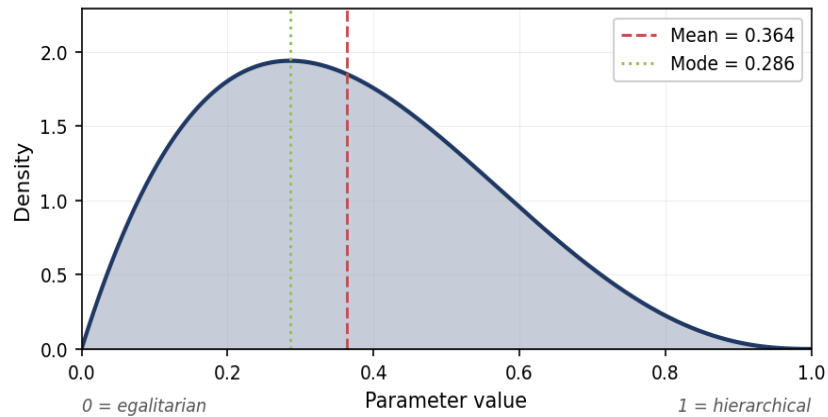
6. Procedural Dependence — Beta(2.5, 2.0)



3.3.7 Tolerance for Asymmetry — Beta(2.0, 3.5)

Proxy: Social Dominance Orientation (SDO7; Pratto et al., 1994; Ho et al., 2015). This is the best-empirically-anchored parameter. The US population mean is 2.98 on a 7-point scale (SD = 1.19). Cross-nationally, means range from approximately 2.5 (Belgium) to 4.0 (UK, Serbia), with all distributions positively skewed. A New Zealand study found 91% of the population in the low-to-moderate range. Rescaled to [0, 1], the US mean of $2.98/7 \approx 0.43$, with strong right-skew matching Beta(2.0, 3.5), which yields a mean of 0.36 reflecting the egalitarian lean of Western democratic populations. If political polarisation produces bimodality, a mixture model (e.g. $0.6 \times \text{Beta}(2, 4) + 0.4 \times \text{Beta}(4, 2)$) may be substituted.

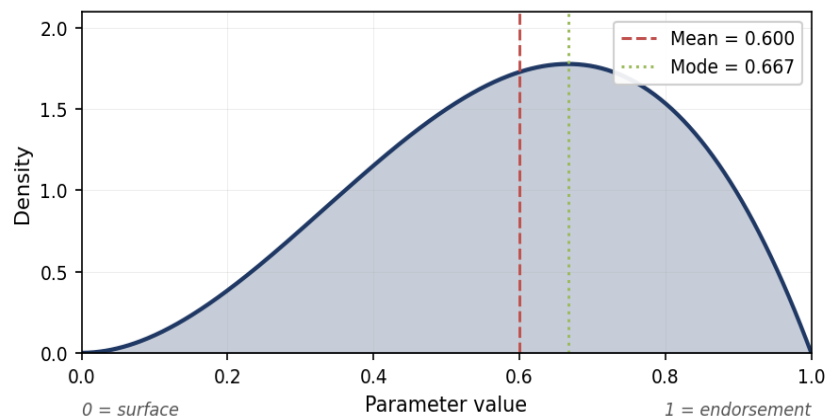
7. Tolerance for Asymmetry — Beta(2.0, 3.5)



3.3.8 Internalisation Dependence — Beta(3.0, 2.0)

Proxy: Self-Regulation Questionnaires (SRQ; Ryan & Connell, 1989). SDT literature consistently shows that identified and integrated regulation (deeper internalisation) are more common than external or introjected regulation in Western adult samples. Autonomy-supportive environments produce better outcomes precisely because most Western adults default to requiring genuine endorsement rather than surface compliance.

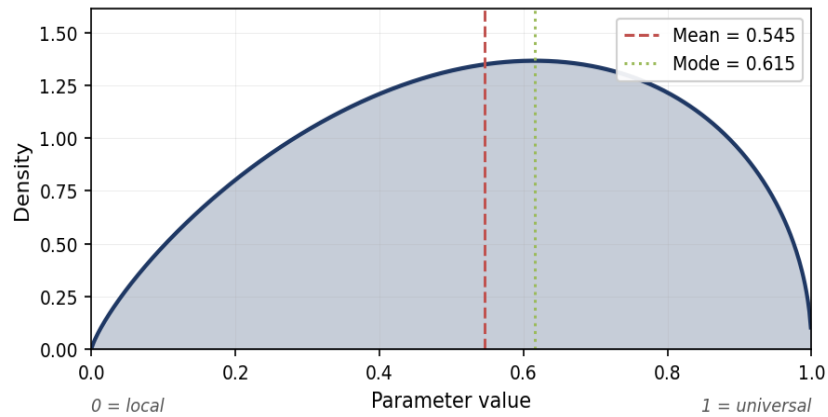
8. Internalization Dependence — Beta(3.0, 2.0)



3.3.9 Moral Scope — Beta(1.8, 1.5)

Proxy: Moral Expansiveness Scale (MES; Crimston et al. 2016); secondary: Moral Foundations Questionnaire (MFQ-2; Atari et al. 2023) individualising/binding ratio and Identification with All Humanity (IWAH; McFarland et al., 2012). The MES is the primary anchor because it is the direct instrument for expanding-circle measurement. Average politically moderate Americans score approximately 20.2–20.5 on MFQ Care and Fairness versus 12.6–16.5 on Loyalty, Authority, and Sanctity. However, the liberal/conservative split produces genuine bimodality. The low-concentration Beta(1.8, 1.5) captures this high-variance, slightly universal-leaning distribution. A Beta mixture model ($0.55 \times \text{Beta}(3, 1.5) + 0.45 \times \text{Beta}(1.5, 3)$) would better capture the political polarisation if desired.

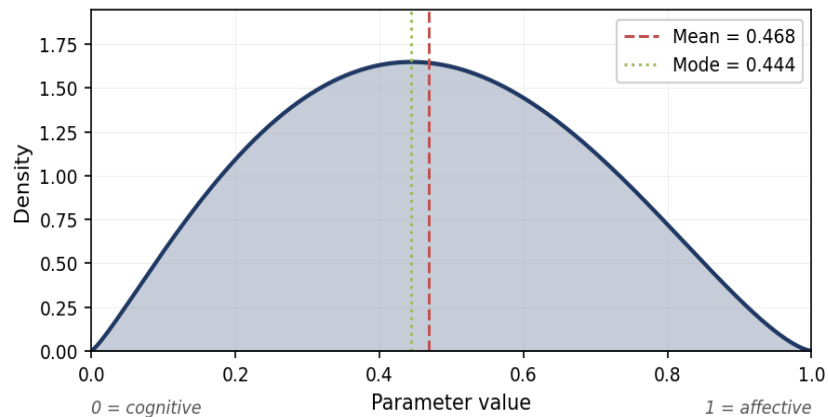
9. Moral Scope — Beta(1.8, 1.5)



3.3.10 Affective Weighting — Beta(2.2, 2.5)

New parameter replacing Temporal Orientation. Proxy: Davis Interpersonal Reactivity Index — Empathic Concern vs. Perspective Taking subscale ratio (Davis, 1983). The IRI has been validated extensively and is cited over ten thousand times. The mild cognitive-leaning (mean ≈ 0.47) reflects the slight cognitive dominance in Western adult samples on reasoned-vs-intuitive moral tasks (Greene et al., 2001; Paxton & Greene, 2010). This parameter is newly introduced in Draft 0.5 and its Beta calibration is preliminary, to be refined against IRI population norms once implementation is underway.

10. Affective Weighting — Beta(2.2, 2.5)



All distribution hypotheses above are starting points to be refined by actual data from the target population. The Beta family handles all of them, and the α/β parameters can be adjusted once empirical calibration data is obtained. A critical methodological concern remains Western bias in source instruments. SDT has demonstrated reasonable cross-cultural validity (Chirkov et al., 2003), but the MFQ has well-documented factor structure problems outside WEIRD populations. The limited population scope is a deliberate response to this concern.

3.4 Joint distribution and dependency structure

Sampling the ten parameters independently from their marginal Betas would generate agents that cannot exist in reality — for example, agents with simultaneously high Constraint Sensitivity and high Tolerance for Asymmetry, a combination that is psychologically incoherent but would appear at non-trivial rates under independent sampling. The joint distribution constrains the sampling space to psychologically plausible regions.

3.4.1 Copula selection

The joint distribution is constructed via a Gaussian copula. By Sklar's theorem, any multivariate distribution can be decomposed into its marginals and a copula function encoding the dependency structure. The Gaussian copula is chosen for the proof-of-concept for four reasons: (1) it handles continuous marginals naturally; (2) it is parameterised by a single 10×10 correlation matrix R , which is interpretable and auditable; (3) it preserves the marginal distributions exactly while introducing realistic inter-parameter correlations; (4) it is well-supported in the implementation stack (NumPy, SciPy).

The main limitation of the Gaussian copula is that it assumes asymptotic tail independence — it does not model situations where parameters are more strongly correlated at the extremes than in the middle of their distributions. If, for example, Constraint Sensitivity and Tolerance for Asymmetry are weakly correlated for moderate agents but very strongly negatively correlated among agents at the extremes, the Gaussian copula would underestimate that tail dependence. A t-copula, which adds a single degrees-of-freedom parameter to model heavier tail dependence, is flagged as a future robustness check. More flexible alternatives (vine copulas, empirical copulas) require either substantially more parameters or joint datasets measuring all ten constructs on the same population — neither of which is available for the proof-of-concept.

3.4.2 Sampling procedure

To sample a single agent's parameter vector $x = (x_1, \dots, x_{10})$:

Step 1. Draw $z \sim \text{MVN}(0, R)$ where R is the correlation matrix specified below.

Step 2. Transform each component to uniform: $u_i = \Phi(z_i)$ where Φ is the standard normal CDF.

Step 3. Transform each uniform to the target marginal: $x_i = F_i^{-1}(u_i)$ where F_i is the Beta CDF for parameter i (or the mixture CDF for parameters 7 and 9 if the mixture fallback is used).

This procedure preserves the marginal distributions exactly while introducing realistic inter-parameter correlations through R . The resulting population occupies only psychologically plausible regions of the ten-dimensional parameter space.

The computational implementation is minimal. In Python:

```
z = numpy.random.multivariate_normal(mean=numpy.zeros(10), cov=R, size=N) u =
scipy.stats.norm.cdf(z) x = numpy.array([scipy.stats.beta.ppf(u[:, i], a[i], b[i])
for i in range(10)]).T
```

where N is the number of agents, R is the correlation matrix, and $a[i]$, $b[i]$ are the Beta shape parameters for parameter i . Three lines of code for the entire joint sampling — consistent with the lean-stack philosophy.

3.4.3 Population size

Each simulation run requires enough agents per societal configuration to detect behavioural differences. With ten parameters and a proof-of-concept subset of societal configurations, a minimum of 200–500 agents per configuration run is necessary for stable emergent dynamics. Multiple runs per configuration (minimum 20) are required for robustness assessment. The full configuration space contains $2^5 = 32$ configurations; the proof-of-concept tests a defensible subset of 8–12 configurations spanning the most antagonistic combinations. The sampling load per full experiment is therefore approximately 10 configurations × 500 agents × 20 runs = 100,000 agent-draws, which is computationally trivial for copula sampling (under one second on commodity hardware).

3.4.4 Correlation matrix R

The matrix R has been verified as positive semi-definite via eigendecomposition (minimum eigenvalue = 0.311). Values are derived from cross-instrument correlation studies in the empirical literature. The correlation matrix is presented in Table 2, with correlations $|r| \geq 0.30$ in bold.

Table 2. Gaussian copula correlation matrix R .

	LL	CS	RT	MoR	RE	PD	TfA	ID	MS	AW
LL	1.00	0.30	0.10	-0.10	-0.25	0.15	-0.20	0.45	0.20	0.05

	LL	CS	RT	MoR	RE	PD	TfA	ID	MS	AW
CS	0.30	1.00	0.25	0.30	-0.30	0.05	-0.35	0.20	0.15	0.25
RT	0.10	0.25	1.00	0.20	-0.10	0.15	-0.30	0.15	0.35	0.25
MoR	-0.10	0.30	0.20	1.00	-0.15	-0.10	-0.10	0.05	0.10	0.15
RE	-0.25	-0.30	-0.10	-0.15	1.00	-0.10	0.25	-0.20	-0.15	0.25
PD	0.15	0.05	0.15	-0.10	-0.10	1.00	-0.10	0.10	0.20	-0.10
TfA	-0.20	-0.35	-0.30	-0.10	0.25	-0.10	1.00	-0.25	-0.40	-0.15
ID	0.45	0.20	0.15	0.05	-0.20	0.10	-0.25	1.00	0.25	0.25
MS	0.20	0.15	0.35	0.10	-0.15	0.20	-0.40	0.25	1.00	-0.15
AW	0.05	0.25	0.25	0.15	0.25	-0.10	-0.15	0.25	-0.15	1.00

3.4.5 Key empirical anchors for R

LL ↔ ID: $r = 0.45$. The strongest correlation in the matrix. GCOS autonomy orientation and SRQ internalised self-regulation share substantial variance within SDT's motivational framework. People who locate legitimacy internally also require deeper endorsement before complying.

TfA ↔ MS: $r = -0.40$. SDO correlates at approximately $r = -0.42$ with MFQ individualising foundations (Care + Fairness) across multiple Western samples. Higher tolerance for asymmetry predicts narrower moral scope. One of the most robust findings in political moral psychology.

CS ↔ TfA: $r = -0.35$. High trait reactance (HPRS) negatively correlates with tolerance for hierarchy (SDO). People who are sensitive to constraint tend to reject asymmetric social arrangements.

RT ↔ MS: $r = 0.35$. Agents with broader moral scope (universalists) tend to have lower tolerance for moral deviation. The Ultimatum Game literature shows that fairness-sensitive rejectors tend to endorse broader justice principles. This is the least directly empirically anchored of the strong correlations — the value is theoretically motivated but based on shared third-variable relationships rather than direct cross-instrument measurement. It is the entry most likely to require adjustment if the Milgram retrodiction test fails.

CS ↔ RE: $r = -0.30$. Independent self-construal positively correlates with personal reactance ($r \approx 0.25-0.35$). Cross-cultural reactance studies confirm that individualists experience stronger personal-freedom threats.

CS ↔ MoR: $r = 0.30$. Constraint-sensitive agents are more likely to respond through external action rather than internal adjustment — a behavioural signature of the reactance tradition.

LL ↔ CS: $r = 0.30$. GCOS autonomy orientation and HPRS reactance proneness share variance: people who locate legitimacy internally are more sensitive to external constraint.

TfA ↔ RT: $r = -0.30$. Agents who accept asymmetry tolerate more violation before responding; agents who reject asymmetry have lower response thresholds. Connects the anti-hierarchy disposition to the activation side of justice.

The newly-introduced Affective Weighting row contains theoretically motivated correlations rather than directly empirical ones. Its relationships with the other parameters (positive with Constraint Sensitivity, Response Threshold, Mode of Response, Relational Embedding, and Internalization Dependence; negative with Procedural Dependence and Moral Scope) are derived from the conceptual mappings between affective/cognitive processing and each parameter's construct. These correlations are explicit targets for sensitivity analysis and for empirical refinement once Affective Weighting is operationalised against IRI population data.

3.4.6 Sensitivity analysis plan

The weaker correlations ($|r| < 0.25$) in R are theoretically motivated but less empirically pinned. To assess the robustness of the simulation to the exact values of these correlations, a three-regime sensitivity analysis is

planned: (a) the proposed R as specified above; (b) R with all $|r| < 0.25$ set to zero (retaining only the strongly anchored correlations); (c) R with all correlations inflated by 20% (testing whether stronger dependencies change the qualitative results). If the five benchmarks retrodict within target ranges under all three regimes, the exact values of the weaker correlations are not load-bearing. If the results are sensitive to a specific correlation, that entry is prioritised for better empirical grounding. If any inflated regime produces a non-positive-semi-definite matrix, the nearest PSD matrix is obtained via Higham's (2002) alternating projections algorithm.

3.5 Concept silos and auditability

At the encoding stage, the five concepts remain separate. Cross-concept interactions are introduced only in the simulation layer. This preserves interpretability and makes it possible to trace a result back through the full chain: source definition, coding decision, parameter profile, behavioural rule, and observed output.

4. Simulation Design

The parameter profile produced by the encoding layer seeds an agent-based population model. The thesis-level goal is not to predict real societies or to prescribe policy. It is to test a narrower and more defensible claim: that canonical conceptual definitions, once formally encoded, can serve as generative inputs whose parameter differences lead to different behavioural worlds.

If the five concepts — each encoded from a single canonical definition — produce repeatable differences in agent behaviour across runs, then the conceptual layer is doing genuine explanatory work. That is the first threshold the framework must cross.

4.1 Construct injection

Four approaches were identified for giving agents their normative worldview, ranked by expected effectiveness. Tool-based injection: the agent is given a callable tool that runs the algorithm and draws from parameter distributions on demand. System prompt injection: definitions, parameters, and algorithmic logic written directly into the system prompt, serving as the initial baseline before the tool-based approach is ready. Vectorisation and retrieval (RAG): construct definitions embedded and retrieved contextually. Fine-tuning: rejected as primary approach due to overfitting risk and static nature.

The proof-of-concept begins with system-prompt injection as the baseline approach. The prompt structure is specified in the companion Experimental Problems and Simulation Protocol document. Tool-based injection is the target architecture for the subsequent iteration, once the system-prompt baseline is validated against the pilot run.

The experimental design draws methodologically on Agents of Chaos (Shapira & Bau, 2026), which reports an exploratory study of six autonomous LLM-powered agents deployed in a live environment with persistent memory, email, Discord access, file systems, and shell execution. Its methodology — particularly around deploying agents to interact with each other and measuring emergent dynamics — serves as a template for the simulation's interaction protocols. A key finding directly relevant here is that agents treat authority as conversationally constructed: whoever speaks with sufficient confidence, context, or persistence can shift the agent's understanding of who is in charge. This reinforces the need for explicit authority encoding rather than relying on emergent role recognition.

4.2 Parameter-to-behaviour mapping

Each parameter maps to a small set of explicit behavioural tendencies defined in advance. This is the methodological hinge of the project: conceptual scores are tied to interpretable mechanisms inside the simulation, rather than translated through vague narrative framing. Legitimacy Locus shifts whether agents accept directives on institutional grounds or require personal endorsement. Constraint Sensitivity determines how readily agents interpret social influence as coercive. Response Threshold sets the violation magnitude required to trigger response. Mode of Response determines whether activation produces internal adjustment or external confrontation. Relational Embedding modulates whether standards are applied uniformly or contextually by relationship. Procedural Dependence shifts evaluation from outcome-assessment to process-assessment.

Tolerance for Asymmetry determines comfort within hierarchical systems. Internalization Dependence determines whether compliance without belief is treated as sufficient or hollow. Moral Scope determines the breadth of moral obligation. Affective Weighting determines whether processing is cognitive-deliberative or affective-intuitive. The full parameter-to-concept-by-behaviour mapping is specified in the companion Parameter-to-Concept Mapping Matrix (fifty cells, eighty-four citations).

4.3 Societal configurations

The simulation has two independent sources of variation. The ten parameters generate a diverse agent population — agents who differ in how sensitive they are to constraint, how readily they defer, how broadly they extend moral obligations, and so on. The societal configurations generate different structural environments those agents are placed into. The parameters model the agents. The configurations model the world.

Each configuration is defined by five concept binaries — whether freedom, justice, authority, care, and loyalty are structurally high or low in the society's normative architecture. These define what the simulation environment enforces, permits, and rewards — not what individual agents believe. They are the rules of the game, not the dispositions of the players.

Freedom (societal level). 0 = the institutional environment constrains individual choice. Agents face structured behavioural expectations, limited exit options, restricted information access, and penalties for deviation. Autonomous action is costly — not impossible, but penalised. 1 = the institutional environment permits and protects individual choice. Agents have exit options, informational access, low penalties for deviation, and structural support for autonomous action.

Justice (societal level). 0 = the institutional environment does not enforce proportionality or procedural fairness. Resource distribution follows power, precedent, or arbitrary allocation. There are no formal complaint mechanisms, no transparency in allocation. 1 = the institutional environment enforces or at least enables proportionality and procedural fairness. Allocation follows transparent rules. Complaint mechanisms exist.

Authority (societal level). 0 = the institutional environment does not vest directive power in particular roles or positions. Coordination is horizontal — through negotiation, consensus, or voluntary association. 1 = the institutional environment vests directive power in recognised roles. Certain agents are structurally empowered to prescribe behaviour for others. Chains of command exist. Disobedience is costly.

Care (societal level). 0 = the institutional environment does not systematically protect the welfare of vulnerable members. Welfare falls to private action or chance. 1 = the institutional environment provides structural supports for vulnerable members — welfare, health, safety nets — and enforces norms of concern for others.

Loyalty (societal level). 0 = the institutional environment does not reward in-group fidelity or impose costs for defection from the group. Memberships are instrumental and exit is unconstrained. 1 = the institutional environment rewards demonstrated loyalty and imposes costs for defection. Group boundaries are marked and enforced.

With five binary axes, the configuration space contains $2^5 = 32$ configurations. This is larger than tractable for a proof-of-concept with realistic computational budget. The proof-of-concept therefore tests a defensible subset of 8–12 configurations spanning the most antagonistic combinations, including the Milgram-analogue configuration (freedom low, justice low, authority high, care low, loyalty low) and its inverse, plus configurations that isolate each concept's effect. The full 32-space characterisation is reserved for later work, as is the possible move to continuous societal parameters in $[0, 1]$ rather than binary axes.

The research question becomes: how does the same population of agents behave differently across configurations, and how do different populations behave within the same configuration? The Stanford Prison Experiment, originally proposed as a secondary benchmark, maps onto a configuration transition: agents start in something approximating high-freedom-justice-care (with authority structurally introduced by the prison setup) and the environment pushes them toward the Milgram-analogue. SPE is treated separately in Section 6.2.

4.4 Illustrative agent-environment interactions

Each concept has a single canonical definition, but the parameter distributions drawn from that definition can be weighted differently across a population. The following examples show how varying the emphasis within the same five definitions produces distinct emergent dynamics when combined with different societal configurations. These are hypotheses to be tested, not guaranteed outputs:

Illustration A — population drawn from the low-legitimacy-locus, high-constraint-sensitivity, low-asymmetry-tolerance, high-internalisation-dependence, high-moral-scope tail, placed in a high-authority-only configuration. Expected: costly coordination failure, high defection from authority demands, low obedience rates resembling Milgram's peer-rebellion conditions.

Illustration B — population drawn from the high-legitimacy-locus, high-procedural-dependence, high-moral-scope, low-constraint-sensitivity, high-care region, placed in a high-authority plus high-justice plus high-care configuration. Expected: strong institutional compliance, redistribution pressure through formal channels, high willingness to coordinate through established rules.

Illustration C — population drawn from the high-asymmetry-tolerance, high-relational-embedding, high-affective-weighting, surface-operable-internalisation, high-loyalty region, placed in a configuration transition from high-freedom-justice-care to the Milgram-analogue. Expected: rapid in-group/out-group divergence and role-based behavioural transformation resembling the Stanford Prison Experiment.

5. Experimental Problems

Beyond the validation benchmarks (Section 6), the framework is exercised through a set of six experimental problems designed to measure the differential effect of the encoded normative architecture on decision-making. These are separate from the benchmarks in function: benchmarks validate that the framework reproduces real human-population patterns; experimental problems test whether the encoded architecture produces differentiated behaviour across configurations in decision contexts that have not been extensively studied in the benchmark literature.

The problems are framed in managerial and strategic contexts. This framing is substantive, not cosmetic: managerial decision-making is irreducibly contested, supports dilemmas with genuine 50/50 ambiguity, and connects the framework to established management-science literature on strategic decision-making under uncertainty.

5.1 The 50/50 baseline-calibration requirement

Each experimental problem must produce an approximately 50/50 binary response distribution under RLHF baseline — that is, before any parameter profile is injected. If the base LLM has a default answer under its alignment training, the encoded architecture's effect on the decision cannot be isolated. Problems are calibrated against GPT-5.4-mini (selected for compliance and non-agentic disposition) prior to deployment. The calibration target is 45/55 to 55/45; problems outside 40/60 to 60/40 are rewritten or replaced.

5.2 Simple problems (LPM route)

Three simple binary-decision dilemmas, deployed at high volume (N = 200 agents per configuration) to provide statistical resolution on configuration-level and within-configuration effects.

S1 — The Promotion Decision. Justice-loyalty tension. Head of a department chooses between a longer-tenured, well-integrated but technically solid candidate (A) and a shorter-tenured, technically superior but less integrated candidate (B). Activates Tolerance for Asymmetry, Relational Embedding, Moral Scope, and Procedural Dependence.

S2 — The Quiet Error. Authority-loyalty-care tension. Mid-level manager has discovered a senior colleague's procedural error that did not affect drug safety or efficacy. Reporting has reputational and institutional costs; staying quiet leaves an irregularity in the record. Activates Internalization Dependence, Legitimacy Locus, Tolerance for Asymmetry, Response Threshold, and Moral Scope.

S3 — The Strategic Pivot. Care-justice tension under strategic uncertainty. Startup CEO decides whether to pivot to an adjacent market (requires laying off six employees) or persist with a failing thesis. Activates Mode of Response, Relational Embedding, Moral Scope, Procedural Dependence, Response Threshold. Note: GPT-5.4-mini may baseline toward pivoting due to exposure to startup-failure narratives; the asymmetry will be flipped during Phase 0 calibration if baseline is outside the 40/60 to 60/40 band.

5.3 Complex problems (Agents-of-Chaos route)

Three multi-agent interactive scenarios, deployed at smaller scale (4–6 agents per run, 20 runs per configuration) to capture emergent group dynamics that the binary response format cannot access.

C1 — The Resource Council. Structural dimension of group decision-making. Five agents deliberate over 5 rounds on the allocation of €500,000 across R&D, employee bonuses, community social-responsibility, and operational cost-reduction. Must reach consensus; default on non-consensus is even 25% split. Measures final allocation, time-to-consensus, proposal patterns, coalition formation, blocking behaviour.

C2 — The Restructuring Board. Hierarchical dimension — Milgram-analogue in managerial framing. Six-agent committee, one designated as CEO, proposing a cost-cutting restructuring with layoffs. Five non-CEO agents vote approve/amend/reject over iterative rounds. Measures approval rate, amendment patterns, deference to CEO, whether human cost is surfaced in deliberation.

C3 — The Scientific-Approach Dilemma. Epistemic dimension — direct engagement with the scientific-decision-making literature in entrepreneurship. Four founder-agents decide over 6 rounds whether to continue or pivot, given asymmetric evidence distributed across them. Measures information-sharing rates, updating patterns, treatment of disconfirming evidence, final vote. This problem has potential thesis-contribution upside beyond the proof-of-concept, because the behavioural signatures (honest updating vs. confirmation bias, escalation of commitment) map directly onto documented patterns in the management-science literature.

5.4 Prompting structure and execution protocol

The system prompt structure (parameter values, normative context, task instruction), per-turn user prompts, and the phased execution protocol (Phase 0 baseline calibration, Phase 1 pilot, Phase 2 full simple-problem experiment, Phase 3 complex-problem experiment, Phase 4 analysis, Phase 5 reporting) are fully specified in the companion Experimental Problems and Simulation Protocol document. The phased structure is designed so that architectural assumptions (injection mechanism, parameter-to-behaviour mapping, marginal calibration) are tested against the pilot before the full experiment commits resources.

6. Validation and Benchmarks

6.1 Performance metric

The simulation's output is a composite welfare score after N time steps. "Performance" is tentatively defined as the rate of morally good decisions made by agents — but this metric is not yet formalised and is flagged as the hardest open question in the project. The operationalisation of moral performance is identified as the critical bottleneck: without a defensible performance metric, the evaluation layer cannot produce meaningful results. Candidate operationalisations include welfare composite scores, norm-deviation metrics, or dimension-specific criteria derived from the benchmark or experimental-problem outputs themselves. The decision is deferred until the pilot phase provides calibration data.

6.2 Benchmarks — one per concept

The framework is validated against five behavioural benchmarks drawn from canonical social psychology, one per concept. Each benchmark provides a specific quantitative retrodiction target: a simulated population placed in the configuration analogous to the benchmark's conditions must reproduce the documented effect size, or the distributions require recalibration. Each benchmark pairs a canonical experimental paradigm with a meta-analytic synthesis that anchors the retrodiction target in population-level data.

6.2.1 Authority — Milgram obedience paradigm

Primary source: Milgram (1974). Twenty-three conditions, 636 participants, 65% baseline maximum-voltage obedience. Meta-analytic anchor: Haslam, Loughnan & Perry (2014) meta-synthesis confirms baseline rate and condition-wise modulator structure. Burger's (2009) partial replication on adjusted protocol produces comparable rates. Retrodiction target: 61–66% maximum-voltage obedience in the authority-high configuration, with the 23-condition modulator structure as secondary targets (obedience drops to ~20% under experimenter absence, rises to ~92% under diffused responsibility, drops to ~10% under peer rebellion). Primary parameters tested: Legitimacy Locus, Internalization Dependence, Tolerance for Asymmetry, Constraint Sensitivity.

6.2.2 Loyalty — Asch conformity paradigm

Primary source: Asch (1956). Original perceptual-judgment paradigm, preserved in its original form rather than re-framed in managerial terms, to maintain direct calibration against the published effect-size literature. Meta-analytic anchor: Bond & Smith (1996), 133 studies across 17 countries, documenting stable baseline conformity and condition-wise modulators. Retrodiction target: approximately 32–37% conformity on critical trials in the loyalty-high configuration. Modulator targets: unanimity-breaking drops conformity to 5–10%; private response drops it to 10–15%; group size increases conformity up to $N = 3$ then plateaus. Primary parameters tested: Relational Embedding, Internalization Dependence, Legitimacy Locus, Tolerance for Asymmetry.

6.2.3 Justice — Ultimatum Game

Primary source: Güth, Schmittberger & Schwarze (1982). Dual-role structure — proposer offers a division, responder accepts or rejects. Rejection leaves both parties with nothing. Meta-analytic anchor: Oosterbeek, Sloof & van de Kuilen (2004), Western-sample meta-analysis; Wallace et al. (2007) twin study (>40% additive genetic heritability of responder threshold). Retrodiction target: Western-calibrated populations produce proposer offers clustering around 40–50% of stake; rejection rate for 20% offers approximately 40–50%; rejection rate for 50% offers approximately zero; responder threshold mean approximately 33% of stake. Cross-cultural variance (Henrich et al. 2005) is reserved for robustness testing and not used for primary retrodiction. Primary parameters tested: Response Threshold, Tolerance for Asymmetry, Moral Scope, Procedural Dependence.

6.2.4 Care — Bystander helping paradigm

Primary source: Latané & Darley (1968, 1970). Subjects observe staged emergencies alone or with confederates; helping rate measured as a function of bystander presence. Meta-analytic anchor: Fischer et al. (2011), 53 studies, 7,700 participants. Retrodiction target: helping rate approximately 75% in alone-condition in the care-high configuration, dropping to approximately 55% with three or more bystanders. Under explicitly dangerous conditions, the bystander-effect attenuation shrinks or reverses. Care-low configurations should produce lower baseline helping regardless of bystander count. Primary parameters tested: Moral Scope, Response Threshold, Relational Embedding, Affective Weighting.

6.2.5 Freedom — Reactance restoration paradigm

Primary source: Worchel & Brehm (1970). Participants rank options; one option is then removed. The attractiveness of options adjacent to the removed one increases measurably (the reactance boomerang), while unrelated options remain stable. Meta-analytic anchor: Rains (2013), meta-analysis of 123 studies on psychological reactance in persuasion contexts. Retrodiction target: option-attractiveness shift of approximately 15–25% following removal, with Cohen's $d \approx 0.45$ per the meta-analytic synthesis. Freedom-low + authority-high configurations should produce stronger boomerang effects; freedom-high configurations should dampen them toward baseline. Primary parameters tested: Constraint Sensitivity, Mode of Response, Legitimacy Locus. This benchmark has a smaller effect size than the other four, and the outcome measure (attitude shift) is noisier than the action measures used elsewhere. The compensating factor: reactance is the appropriate per-concept validation anchor for freedom, and the weakness is in noise rather than in construct validity.

6.2.6 The Stanford Prison Experiment, considered and rejected

An earlier iteration of this framework included the Stanford Prison Experiment (Zimbardo, 1971) as a secondary benchmark. The SPE is rejected on substantive grounds in the present specification. The methodological critique is extensive: Le Texier (2018) documented direct experimenter coaching of guards, Carnahan & McFarland (2007)

documented self-selection bias, and Haslam & Reicher's work (including the BBC Prison Study, 2006) has contested the conformity-to-role interpretation on empirical grounds. More decisively from a methodological standpoint, the SPE provides no quantified retrodiction target: "guards became abusive" is a qualitative pattern, not a numerical benchmark, and cannot be used for simulation validation in the way Milgram's 65% baseline or Asch's 35% critical-trial rate can. The role-transformation phenomenon is nevertheless relevant to the framework; it is addressed at the experimental-problem layer via C2 (The Restructuring Board), which provides a structurally analogous scenario in managerial framing with quantifiable behavioural output.

6.3 Distribution validation

The empirically calibrated distributions provide three distribution-specific validation axes beyond the benchmark retrodiction:

Internal consistency. Agents with extreme parameter combinations that are psychologically implausible should be rare in the sampled population. The correlation structure in R ensures this. Frequency analysis should confirm that fewer than 2% of agents occupy jointly extreme positions on negatively correlated parameters.

Configuration sensitivity. Populations drawn from each configuration should show the expected direction of behavioural shift across benchmark tasks. The Milgram-analogue configuration should produce the Milgram retrodiction; its inverse should produce the lowest obedience; the loyalty-high configuration should produce the Asch retrodiction; and so on.

Sensitivity to R. The three-regime sensitivity plan (as-specified, weak-entries-zeroed, all-inflated-20%) is run on all five benchmarks. A benchmark that passes only under one regime is flagged as sensitive to correlation structure; one that passes across all three is architecturally robust.

6.4 General validation axes

Validation proceeds along five axes. None is sufficient alone; together they constitute the minimum standard for the proof-of-concept to be considered credible.

Experimental replication. Emergent behaviour in benchmark scenarios should be structurally comparable to documented experimental patterns, within the retrodiction bands specified above.

Face validity. Canonical cases should land in plausible regions of the parameter space.

Mechanistic validity. Every parameter must have a clear, documentable behavioural consequence inside the simulation — the parameter-to-behaviour mapping cannot be post-hoc.

Robustness. Observed differences must persist across random seeds and uncertainty draws rather than appearing in a single attractive run.

Interpretability. Every output must remain traceable from observed dynamics back through the behavioural rule, the parameter profile, the coding decision, and the source definition.

7. Implementation

The implementation relies on a deliberately lean stack. Each component is chosen because it addresses a specific layer of the pipeline; none is load-bearing beyond its designated role. Bayesian encoding: PyMC for priors, posteriors, and uncertainty-aware scoring of the five canonical definitions. Network logic: pgmpy only if an explicit Bayesian network proves necessary; otherwise the first version stays PyMC-first. Data processing: Python, pandas, and NumPy for coding tables, aggregation, metric computation, and reproducible analysis. Population model: Mesa for the agent-based testbed and repeated simulation runs across contrasting population configurations. Visualisation: Plotly for a comparative dashboard. Joint distribution sampling: NumPy's multivariate_normal for the Gaussian copula draws, SciPy's norm.cdf and beta.ppf for the marginal transformations. Injection: system-prompt baseline initially, tool-based mechanism as follow-up. Agents of Chaos orchestration code and LPM-based simulation code are available in prior form and will be adapted for parameter injection.

8. Phased Work Plan

The project is structured into six phases. Each phase produces a concrete deliverable that the subsequent phase depends on.

Phase 1 — Foundations. Finalise the shared parameter set across all five concepts. Define parameter endpoints. Abstract freedom, justice, authority, care, and loyalty each to a single canonical definition grounded in the psychological literature. Status: complete. Pentad locked under the ≥ 3 -of-4 convergence criterion. Ten parameters defined with explicit endpoints. Five canonical definitions established with source literature. Universality validated through the companion mapping matrix (50 cells, 84 citations).

Phase 2 — Distribution modelling. Assign statistical distributions per parameter, grounded in empirical data on human populations. Select population scope. Validate the plausibility of chosen distribution families against available survey data. Construct the joint distribution via Gaussian copula with empirically anchored correlation matrix. Status: substantially complete. Ten marginal Beta distributions specified with proxy instruments. Correlation matrix R constructed and verified positive semi-definite (min eigenvalue 0.311). Five per-concept benchmarks with retrodiction targets specified. Sensitivity analysis plan specified.

Phase 3 — System architecture. Build the tool-based injection mechanism as the target approach. Prototype the system-prompt baseline for Phase 0 calibration and Phase 1 pilot. Define the agent decision interface through which parameterised constructs inform agent behaviour.

Phase 4 — Pilot and calibration. Phase 0 baseline-calibrate the six experimental problems against GPT-5.4-mini. Phase 1 pilot run on one simple problem under one configuration to verify that the encoded architecture produces differentiated behaviour, tractable to parameter values rather than noise.

Phase 5 — Full simulation. Implement benchmark experiments (five per-concept benchmarks) and run the full experimental-problem protocol (simple at $N = 200$ per configuration, complex at 20 runs per configuration). Apply the three-regime R-sensitivity analysis across all runs.

Phase 6 — Analysis and writing. Compare population configurations, highlight behavioural differences across the encoded concepts, and frame the results as a proof-of-concept. Write the thesis as a methods paper with clearly bounded claims. Candidate thesis-contribution findings (particularly from C3 on scientific decision-making) are written up for potential separate venues.

9. Honest Assessment: What Is and Is Not Established

9.1 What is conceptually solid

The reframing of normative encoding as a structural problem rather than a compliance problem. The choice of freedom, justice, authority, care, and loyalty as the pentad under a falsifiable ≥ 3 -of-4 convergence criterion across Rokeach, Duckitt-Sibley, Schwartz, and MFT. The single-canonical-definition approach grounded in the psychological rather than purely philosophical literature. The shared ten-parameter architecture validated for universality through the companion matrix. The empirically calibrated joint distribution with Beta marginals and Gaussian copula, grounded in instrument-level population data from SDO, GCOS, HPRS, MFQ/MES, SCS, UG, STAXI, SRQ, Colquitt, Davis IRI, and identity-fusion scales. The binary societal configurations as independent structural variation orthogonal to agent-level parameters. The per-concept benchmark architecture — one behavioural anchor per concept with meta-analytic retrodiction target. The substantive rejection of SPE on documented methodological grounds with quantified-target unavailability as the decisive factor. The sensitivity analysis plan for the correlation matrix. The three-layer separation of encoding, simulation, and validation. The explicit 50/50 baseline-calibration protocol for experimental problems, which allows the architecture's effect to be isolated causally.

9.2 What is not yet established

A working definition of "morally good decisions" in measurable terms. A validated tool-based injection mechanism (system-prompt baseline is first, tool-based is target). Any empirical validation of the framework against actual

agent simulation data. The minimum population size required for meaningful emergent dynamics (200–500 per configuration is hypothesised but untested). Whether static distributions are sufficient or dynamic drift is necessary. The precise behavioural consequences of each parameter inside the simulation — these are proposed in the mapping matrix but not yet behaviourally tested. Whether the ten-parameter set is empirically redundant despite conceptual distinctness (to be tested via exploratory factor analysis once initial encodings are complete). Whether the proposed correlation matrix R accurately captures the true dependency structure — the weaker correlations ($|r| < 0.25$) are theoretically motivated but not yet empirically validated, and the $RT \leftrightarrow MS$ correlation ($r = 0.35$) is the least directly anchored of the strong correlations. Whether the Gaussian copula's tail independence assumption is adequate or whether heavier-tailed alternatives (t-copula) would change the qualitative simulation results. Whether the Affective Weighting parameter's Beta calibration and R -row values, both of which are preliminary in Draft 0.5, are empirically correct once the IRI calibration data is incorporated. Whether the $2^5 = 32$ configuration space can be adequately sampled by the 8–12-configuration proof-of-concept subset or whether missing configurations produce important behavioural regimes that the proof-of-concept misses. Whether the freedom benchmark's noisier outcome measure produces enough signal-to-noise for retrodiction to discriminate configurations at the same resolution as the other four benchmarks.

10. Open Questions

The following questions are not footnotes. They are load-bearing decisions that the downstream phases depend on. Each must be resolved before the evaluation layer can produce meaningful results. What is the right granularity for "morally good"? Binary, scalar, or multi-dimensional? Should distributions be static for a simulation run, or should they drift as agents interact? How many agents constitute a population sufficient to generate meaningful welfare scores? Beyond Milgram and Asch, which classic experiments offer the best combination of replicability and moral complexity for complementary validation? Is the tool-based injection mechanism sufficient to preserve agent flexibility, or does the indirection introduce artefacts that distort the conceptual encoding? Are any of the ten parameters empirically correlated despite conceptual independence, and does the proposed R capture these dependencies adequately or miss non-linear interactions? How should the Agents of Chaos findings on conversationally constructed authority be incorporated — as a parameter modulator, a simulation-layer rule, or a distinct injection mechanism? Should the societal configurations remain binary, or should they be replaced by continuous societal parameters in later phases? Does the three-regime sensitivity analysis for R adequately assess the robustness of the joint distribution, or are additional perturbation strategies needed? What is the appropriate subset of the 32 configurations for the proof-of-concept, and by what criteria is that subset to be defended?

11. Scope and Future Direction

The enforced initial scope is deliberately narrow: five political-ethical concepts, ten shared parameters, one joint distribution, a defensible subset of the 32-configuration space, six experimental problems split between simple and complex, and five per-concept benchmarks. Expansion to additional concepts or more demanding applications happens only after this core proves out. This constraint applies throughout the project lifecycle.

The framework is conceived as a training ground for a larger research programme: AI systems whose normative interpretation is shaped by an intrinsic conceptual architecture rather than external rule patches. The thesis does not claim to solve alignment or produce a fully ethical artificial agent. Its immediate contribution, if successful, is to demonstrate that stable, explicit conceptual structures can be formalised, compared, and used to generate distinct judgment patterns in a controlled setting.

If the proof-of-concept succeeds, the next step is to investigate whether such structures can condition AI decision-making over time without collapsing into rigidity, ideology, or mere persona. That investigation is future work and is not claimed or promised by this document. Related open extensions include the move to continuous societal parameters, the incorporation of dynamic parameter drift during simulation, the expansion of the concept pentad to include candidates (tradition, sanctity, security) that did not clear the ≥ 3 -of-4 convergence bar, and the generalisation of the framework beyond Western-democracy calibration to other cultural contexts — the last of which would require substantial re-validation of every proxy instrument used in the calibration.

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